LAB – Inequality

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Instructions, Part 1

- 1. I have provided code on how to load and use the World Inequality Database. Use it to:
  - a. Display the Lorenz Curve for Sweden and another (European) country of your choice.
  - b. Calculate the Gini Coefficient for Sweden and another (European) country of your choice using the formula:

    $$\text{Gini} = \frac{2}{\bar{x}} \cdot \text{COV}(x_i, R_i)$$

    where x_i is the income for individual i and R_i is the income rank of individual i.
  - c. Calculate the Gini Coefficient for Sweden and another (European) country of your choice using the `Ineq` package.
  - d. Compare and analyze the results.

Instructions, Part 2

- 1. I have provided data (ESS) and code to decompose Swedish inequality into a variance decomposition.
  - a. Run the script to get the effect of years of education and father's years of education on household income. This is the regression estimated (see last slide for what the parameters means):

    $$y_i = \alpha + \beta x + \delta z + \epsilon$$

  - b. Provide the results of the variance decomposition. How much of the variance is explained by years of education, father's years of education, and the covariance between the two? You do not need to report the results of the error terms (they are excluded from the decomposition on the net slide anyway)
  - c. Do the same decomposition but for another country in the ESS data. Compare and analyze the results.
  - d. Write a lab report. Motivate why your choice of country and analyze the results (300-500 words).

Variance Decomposition

Variance decomposition of education and parent years of education:

$$\text{Var}(y_i) = \beta^2\text{Var}(x) + \gamma^2\text{Var}(z) + 2\beta\gamma\text{Cov}(x, z)$$

- y_i: Household Income
- x: Years of Education.
- z: Father Years of Education
- β: Effect of Years of education.
- δ: Effect of Father years of education.
- ϵ: Error term